Assignment - 3

(Code is Below the Question)

Take a list of numbers as input and print the largest and smallest numbers in the list.
Take a string as input and print the length of the string.
Take a list of names as input and print the list in alphabetical order.
Take a list of numbers as input and print the total sum of the elements in the list.
Take a string as input and print the string in uppercase.

Write a user-defined function factorial(n) that takes a positive integer n as an argument and returns its factorial. Use the function in a program and print the factorial of a given number.

Write a user-defined function "find_largest(a, b, c)" that takes three numbers as arguments and returns the largest of the three. Use the function in a program to find and print the largest of three given numbers.

Write a user-defined function greet(name) that takes a name as an argument and prints a greeting message like "Hello, [name]!" Use the function to greet the user with their name.

Combining Built-in and User-Defined Functions: Find the Average of a List: Write a user-defined function average(numbers) that takes a list of numbers as an argument and returns the average of the numbers. Call the function with a list of numbers and print the average.

Check Palindrome: Write a user-defined function is_palindrome(s) that takes a string as an argument and returns True if the string is a palindrome (reads the same forward and backward), and False otherwise. Test the function with different strings and print the results.

Use the Math Module:
Write a program that imports the math module and uses it to: Find the square root of a number.
Calculate the sine of an angle .
Find the greatest common divisor (GCD) of two numbers .

Use the Random Module:
Write a program that imports the random module and uses it to: Generate and print a random integer between 1 and 100.
Create a list of random numbers and print the list.
Shuffle a list of numbers and print the shuffled list.

Use the Datetime Module:
Write a program that imports the datetime module and uses it to: Get and print the current date and time .
Calculate and print the number of days between two dates.
Format and print the current date in the format "DD-MM-YYYY"

Use the OS Module:
Write a program that imports the os module and uses it to: Print the current working directory
Create a new directory and verify its existence.
List all files and directories in the current directory

Create and Use a Custom Package:
Create a package called my_math with two modules: arithmetic.py and geometry.py. In arithmetic.py, define functions for addition, subtraction, multiplication, and division. In geometry.py, define functions to calculate the area of a circle and the area of a rectangle.

Write a program that imports these functions from the my_math package and uses them to perform various calculations.

Create a Package for String Utilities:
Create a package called string_utils with two modules: string_operations.py and string_validations.py. In string_operations.py, define functions for reversing a string, converting a string to uppercase, and finding the length of a string. In string_validations.py, define functions to check if a string is a palindrome and if it contains only alphabetic characters.

Write a program that imports and uses these functions from the string_utils package.

Code:

```
# Assignment: Day 3 - Using Built-in Functions, User defined functions,
Built-in Modules, and User-defined Modules

import math
import random
import datetime
import os
from my_math.arithmetics import add, subtract, multiply, divide
from my_math.geometry import area_of_circle, area_of_rectangle
from string_utis.string_operations import reverse_string, to_uppercase,
get_length
from string_utis.string_validation import is_only_alpha, is_only_alpha

# 1. Using Built-in Functions

def built_in_functions_demo():
```

```
    # Max and Min functions
    nums = list(map(int, input("Enter a numbers with spaces: ").split()))
    print("The largest number is:", max(nums))
    print("The smallest number is:", min(nums))

    # String length - len function
    string = input("\nEnter a string: ")
    print('The length of the string is: ', len(string))

    # Alphabetical order - sorted function
    names = input("\nEnter names separated by space: ").split()
    names.sort()
    print("Names in alphabetical order:", names)

    # Sum of list - sum function
    numbers = list(map(int, input("\nEnter numbers separated by space:
").split()))
    print("The sum of the numbers is:", sum(numbers))

    # Uppercase
    str = input("\nEnter a string: ")
    print('The string in upper case is: ', str.upper())


# 2. User defined functions

# Factorial
def factorial(n):
    result = 1
    for i in range(1, n + 1):
        result = result * i
    return result



# Find Largest of Three
def find_largest(a, b, c):
    if a >= b and a >= c:
        return a
    elif b >= a and b >= c:
```

```python
        return b
    else:
        return c


# Greeting
def greet(name):
    print("Hello, " + name + "!")



# Average
def average(numbers_list):
    total = sum(numbers_list)
    count = len(numbers_list)
    return total / count

my_nums = [10, 20, 30, 40]


# Palindrome
def is_palindrome(s):
    s = s.lower()
    reverse_s = s[::-1] # This flips the string
    return s == reverse_s




# 3. MODULES - Built-in

def modules_demo():

    # Math Module
    print("\nUsing math module:")
    print("\nSquare root of 64:", math.sqrt(64))
    print("Sine of 90 degrees:", math.sin(math.radians(90)))
    print("GCD of 48 and 60:", math.gcd(48, 60))

    # Random Module
```

```python
    print("\nUsing random module:")
    print("\nRandom number (1-100):", random.randint(1, 100))
    random_list = [random.randint(1, 10) for i in range(5)]
    print("Random list:", random_list)
    random.shuffle(random_list)
    print("Shuffled list:", random_list)

    # Datetime Module
    print("\nUsing datetime module:")
    now = datetime.datetime.now()
    print("\nCurrent date/time:", now)
    print("Formatted date:", now.strftime("%d-%m-%Y"))

    # OS Module
    print("\nUsing os module:")
    print("\nCurrent folder:", os.getcwd())
    # Creating a folder safely
    if not os.path.exists("new_folder"):
        os.mkdir("new_folder")
        print("Folder created!")
    print("Items in directory:", os.listdir())



# EXECUTION
if __name__ == "__main__":
    print('\nUsing Built-in Functions')
    built_in_functions_demo()

    print('\nUsing User defined functions')
    print("\nFactorial of 5:", factorial(5))
    print("Largest of 10, 50, 30 is:", find_largest(10, 50, 30))
    greet("Deekshita")
    print("Average of list:", average(my_nums))
    print("Is 'madam' a palindrome?:", is_palindrome("madam"))

    print('\nUsing Built-in Modules')
    modules_demo()

    print('\nUsing User-defined Math Modules')
```

```
    print("\nArea of circle with radius 5:", area_of_circle(5))
    print("Area of rectangle with length 10 and width 5:",
area_of_rectangle(10, 5))
    print("Addition of 10 and 5:", add(10, 5))
    print("Subtraction of 10 and 5:", subtract(10, 5))
    print("Multiplication of 10 and 5:", multiply(10, 5))
    print("Division of 10 and 5:", divide(10, 5))
    print("Division of 10 and 0:", divide(10, 0))

    print('\nUsing User-defined String Modules')
    print("\nReverse of 'Hello World':", reverse_string("Hello World"))
    print("Uppercase of 'hello world':", to_uppercase("hello world"))
    print("Length of 'Hello World':", get_length("Hello World"))
    print("Is 'Hello' only alphabets?:", is_only_alpha("Hello"))
    print("Is 'Hello123' only alphabets?:", is_only_alpha("Hello123"))
```

Output:

C:\Users\Dimple\OneDrive\Desktop\Working Directory\Wipro_Assignments>C:/Users/Dimple/AppData/Local/Programs/Python/Python312/python.exe "c:/Users/Dimple/OneDrive/Desktop/Working Directory/Wipro_Assignments/day-3/day3.py"

Using Built-in Functions
Enter a numbers with spaces: 4 3 6 2 5 1 9
The largest number is: 9
The smallest number is: 1

Enter a string: Hello Wipro!!
The length of the string is: 13

Enter names separated by space: Riya Babitha Ronak Piyush
Names in alphabetical order: ['Babitha', 'Piyush', 'Riya', 'Ronak']

Enter numbers separated by space: 4 12 7 8 4 9
The sum of the numbers is: 44

Enter a string: the sun rises in the east

The string in upper case is: THE SUN RISES IN THE EAST

Using User defined functions

Factorial of 5: 120
Largest of 10, 50, 30 is: 50
Hello, Deekshita!
Average of list: 25.0
Is 'madam' a palindrome?: True

Using Built-in Modules

Using math module:

Square root of 64: 8.0
Sine of 90 degrees: 1.0
GCD of 48 and 60: 12

Using random module:

Random number (1-100): 4
Random list: [4, 5, 3, 7, 9]
Shuffled list: [7, 5, 4, 3, 9]

Using datetime module:

Current date/time: 2026-04-27 02:33:01.266890
Formatted date: 27-04-2026

Using os module:

Current folder: C:\Users\Dimple\OneDrive\Desktop\Working Directory\Wipro_Assignments
Items in directory: ['day-1', 'day-3', 'new_folder']

Using User-defined Math Modules

Area of circle with radius 5: 78.53975
Area of rectangle with length 10 and width 5: 50
Addition of 10 and 5: 15
Subtraction of 10 and 5: 5
Multiplication of 10 and 5: 50
Division of 10 and 5: 2.0
Division of 10 and 0: Error: Division by zero is not allowed.

Using User-defined String Modules

Reverse of 'Hello World': dlroW olleH
Uppercase of 'hello world': HELLO WORLD
Length of 'Hello World': 11
Is 'Hello' only alphabets?: True
Is 'Hello123' only alphabets?: False

arithmetics.py file

```python
# Arithmetic functions
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Error: Division by zero is not allowed."
    return a / b
```

geometry.py

```python
# Area of Circle and Rectangle using User-defined Functions and Modules

def area_of_circle(radius):
    pi = 3.14159
    return pi * radius ** 2

def area_of_rectangle(length, width):
    return length * width
```

sting_operations.py

```python
# Function to reverse a string
def reverse_string(s):
    return s[::-1]

# Function to convert to uppercase
def to_uppercase(s):
    return s.upper()

# Function to find the length
def get_length(s):
    return len(s)
```

string_validations.py

```python
# Function to check if it reads the same backward
def is_palindrome(s):
    s = s.lower()
    return s == s[::-1]

# Function to check if there are only letters (no numbers or symbols)
def is_only_alpha(s):
    return s.isalpha()
```